ESAS AUGUST 2019

1. What can you say about the following statement regarding centrifugal pumps? The discharge of the centrifugal pumps can be throlled without the building up of an excessive pressure or overloading its driving unit
   I. True
   II. False
   A. Either I or II
   B. Both I or II
   C. II
   D. I
2. A description of arrangements of atoms or ions in a material is called_____?
   A. Mixture
   B. Composition
   C. Component
   D. Structure
3. Which of the following is qualified as the chairman of the Continuing Professional Development Council for Electrical Engineering?
   A. Representative from Association of Professional Organization
   B. PRBEE board member
   C. Representative from academy
   D. PRC commissioner
4. Which of the following is not a unit of work?
   A. N-m
   B. Erg
   C. Dyne
   D. Kg-sq.m/sq. sec
5. What is UPS stands for?
   A. Uninterrupting Power Supply
   B. Uninterruptible Power Supply
   C. United Power Supply
   D. Uninterruptible Portable Supply
6. The central processing unit of computer is a combination of _____
   A. Arithmetic logic and input unit
   B. Control and storage
   C. Arithmetic logic and control unit
   D. Control and input unit
7. What is the maximum distance from flooring to the receptacle outlet in dwelling unit in mm?
   A. 2500
   B. 1800
   C. 1500
   D. 2800
8. What refers to the slender member which prevents parts of a structure moving to each other under compressive forces?
   A. Tie
   B. Column

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C. Arch
D. Strut

9. Materials in this group are composed of one or more metallic elements such as iron, aluminum, copper, titanium, gold and nickel. What is this group called?
A. Metals
B. Metalloids
C. Solids
D. Crystals

10. Where receptacles are connected to a 30-A branch circuit, the maximum allowable cord and plug connected load shall not be more than _____A?
A. 24
B. 26
C. 30
D. 34

11. In Rigid Metal Conduit (RMC) wiring, conduits shall be supported at least every _____mm?
A. 1500
B. 2000
C. 3000
D. 1000

12. The unit of joule can be converted to which of the following?
A. Pa-m^2
B. N-kg
C. Pa-m^3
D. Pa/m^2

13. Which of the following computer program is not considered open soure-software?
A. Firefox
B. Microsoft Office
C. Linux
D. OpenOffice.org

14. The total load on overcurrent device located in panelboard shall not exceed a certain percentage of its rating. What is this percentage?
A. 120%
B. 90%
C. 100%
D. 80%

15. In combustion chamber, fuel is burned to raise the temperature of the medium prior to the production of work through an expansion process. The heat generated during the combustion reaction is the _____?
A. Heat of formation
B. Specific heat
C. Heat of vaporization
D. Heat of reaction

16. Which of the following do you called the brain of any computer system?
A. Arithmetic logic unit
B. Central processing unit

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C. Random access memory
D. Control processing unit

17. How may times the diameter of the largest raceway shall have a length of the box be in straight pulls?
A. 6
B. 4
C. 8
D. 10

18. The Philippine Electrical code Part 1 2009 is based on the National Electrical Code 2005 while the Philippine Electrical Code Part 1 2017 is based on the National Electrical Code_____?
A. 2017
B. 2011
C. 2008
D. 2014

19. A printed circuit board that has (at least) slots to connect cards into. Often, they also include a CPU and memory is called____
A. Mainframe
B. IDE port
C. Motherboard
D. Server

20. What is the metal used to recover copper from a solution of copper sulphate?
A. Ag
B. Hg
C. Fe
D. Na

21. Which of the following is/are hazardous practice/s.
A. Using equipment improperly
B. Failure to warn of secure
C. All of these
D. Horseplay

22. What is the reasons way incidents are investigated?
A. To prevent or control a future reoccurrence on a similar incident
B. To determine the cause of incident
C. To gather evidence of the incident
D. To determine who is at fault in the incident

23. Which law that for a confined fluid, the pressure at a point has the same magnitude in all direction?
A. Pascal's Law
B. Bernoulli's law
C. Avogadro's law
D. Amagat Law

24. Which of the following treatments will result in an increase in the fatigue strength of the steel?
A. Shot peening
B. Surface roughening

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C. Cold working
D. Annealing

25. Which of the following is the basic symbol used as a beginning or termination in logic diagram or flowcharts?
A. Square box
B. Oval box
C. Rectangular box
D. Diamond box

26. What is the transfer of heat due to motions of a fluid near the surface of an object?
A. Insulation
B. Radiation
C. Convection
D. Conduction

27. Objects that emit their own light are called _____ bodies?
A. Solar
B. Luminated
C. Photovoltaic
D. illuminated

28. What form of software can also be called firmware?
A. Cache memory
B. Operating memory
C. System BIOS
D. Random memory

29. In a mixed gas such as air, the rms speed of different molecules are _____?
A. Proportional to molecular mass
B. Independent of molecular mass
C. Proportional to the square root of molecular mass
D. Inversely proportional to the square root of molecular mass

30. Under which of the following type of loading does fatigue occur?
A. Repeated load
B. Plane load
C. Static load
D. High load

31. What is the heaviest subatomic particle?
A. Electron
B. Proton
C. Atom
D. Neutron

32. Which of the following insulation color identify the grounded conductor of s branch circuit?
A. Black
B. Green
C. Yellow
D. White

33. What is the maximum number of overcurrent devices for a lighting and appliance that can be installed in a single cabinet?

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A. 36
B. 42
C. 24
D. 48

34. What physical property of material that refers to the point at which a material liquefies on heating or solidifies on cooling
A. Curie point
B. Specific heat
C. Melting point
D. Refractive index

35. What refers to the establishing interrelationship between people and things in such a way that human and materials resources are effectively focused toward achieving the goal of the company?
A. Planning
B. Leading
C. Organizing
D. Controlling

36. Who is the present chairman of the Board of Electrical engineering?
A. Jaime Mendoza
B. Rodolfo Penelosa
C. Francis Mapile ADEL GARCIA
D. Fortunato Leynes

37. Which of the following refers to the parallel axis theorem for second moment area for strength of materials?
A. Maxwell Theorem
B. Young's Theorem
C. Steiner's Theorem
D. Mohr's Theorem

38. The mathematical expression (1+i) raised to n in Engineering Economics is referred to as____?
A. Compound value
B. Interest factor
C. Compound amount factor
D. Future value

39. The 'earth' best defines of which of the following term?
A. Bonding
B. Ground electrode
C. Ground
D. Grounded conductor

40. What is the standardized communication technique for expressing instruction to a computer?
A. Flow chart
B. Programming language
C. Algorithm
D. Logic

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41. Branch circuit conductors supplying a single motor shall have an ampacity of how many percent of its rated full-load current?
    A. 100%
    B. 115%
    C. 150%
    D. 125%
42. An open system of first law of thermodynamics should be utilized for all EXCEPT_____?
    A. A turbine
    B. A piston-cylinder device with no inlet/exhaust valve
    C. A compressor
    D. A nozzle
43. What is the classification of a turbine if the water moves along the vanes toward the axis of rotation of the runner?
    A. Axial Flow
    B. Radial Flow
    C. Parallel Flow
    D. Mixed Flow
44. Which of the following in the thermodynamics is considered as the heat content of a system?
    A. Internal Heat
    B. Molar heat
    C. Entropy
    D. Enthalpy
45. What refers to the point in which bending moment changes sign through a zero value in strength of materials?
    A. Zero stress point
    B. Critical point
    C. Point of contraflexures
    D. Point of inflection
46. Which one of these statements is TRUE for twinning?
    A. It is the most significant form of plastic deformation
    B. It cannot be used by impact or thermal structures
    C. It occurs at lower shear stresses than slip
    D. It frequently occurs in hexagonal closed-packed structure
47. What law of nature underpins the concept of a balanced chemical equation?
    A. Gay-Lussac Law
    B. Faraday's Law
    C. Second law of thermodynamics
    D. First Law of Thermodynamics
48. The shape of the bending moment diagram over the length of beam, having no external load is always_____?
    A. Cubical
    B. Circular
    C. Parabolic
    D. Linear

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49. The clearance between conductors and roof where the service conductors are passing over roofs must be ____mm?
    A. 1000
    B. 1500
    C. 2000
    D. 2500
50. Which of the following portion of hard disk space served as RAM?
    A. Cache
    B. Random Memory
    C. Read-only-memory
    D. Virtual memory
51. The amount of property in which a willing buyer will pay to a willing seller for the property when neither one is under the compulsion to buy nor sell is called ___ value?
    A. Goodwill
    B. Second hand
    C. Market
    D. Fair
52. If you are appointed as a division manager, which of the following task is most likely you will do?
    A. Determine the resource needed
    B. Develop strategies and tactics
    C. Set a standard
    D. Set goals
53. Which of the following Republic Act that provide framework for the restructuring of electric power industry including the privatization of the asset of NPC, the transition to the desired competitive structure
    A. 7920
    B. 9136
    C. 7832
    D. 8901
54. Which of the following best defined the phrase “likely to become energized”
    A. Failure of insulation
    B. High-voltage equipment
    C. Near energized electrical equipment
    D. Equipment that is exposed to lightning
55. In house wiring, ground conductor must not be_____.
    A. Insulated
    B. Identified
    C. Placed in a circuit
    D. Fused
56. What refers to ions that are not involved in the overall reactions?
    A. Guest room
    B. Spectator ions
    C. Extra Ions
    D. Special ions

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57. The following valves are used for shut-off service. Which one is not included?
    A. Plug cock
    B. Ball valve
    C. Butterfly valve
    D. Gate valve
58. Current in an electrical circuit will take _____ to return to the source.
    A. A high impedance path only
    B. The path in the earth
    C. Only the path of least resistance
    D. Any and all paths available
59. The usual function of a disconnect switches in high voltage is to ____.
    A. Open the circuit in the vent of overload
    B. Open or close the circuit under load
    C. Maintain continuity of service
    D. Isolate from energized buses, equipment which are not in service
60. Which of the following anchor is designed for medium and heavy weight fastening in solid materials?
    A. Anchor clamp
    B. Toggle bolt anchor
    C. Wall grip anchor
    D. Stud-bolt anchor
61. Which of the following does not transfer work to or from a system?
    A. An electrical resistance heater
    B. A moving piston
    C. The expanding membrane of balloon
    D. A membrane that burst
62. What is the interest accrued over the interest period is also subject to the interest rate in the next period?
    A. Quarterly interest
    B. Compound interest
    C. Simple interest
    D. Compound interest
63. What is the standard color for safety signages to indicate the presence of radiation hazards example in X-ray laboratory in the hospital?
    A. Green with combination of red
    B. Orange
    C. Blue
    D. Purple with combination of yellow
64. Encryption can provide all of the following features of information systems security except_____.
    A. Confidentiality
    B. Integrity
    C. Authencity
    D. Privacy
65. This provides detailed information about a specific hazardous material and contains some of the following such name of substance physical hazards, health

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hazards, safe handling procedure, control measures, emergency first aid procedures, etc. What is this called?

A. Management safety duty sheet
B. Material Signage data sheet
C. Management safety data sheet
D. Material safety data sheet

66. Which of the following technique must a manager used if he decides to absorb the risk in the project?

A. Use a different technology
B. Buying insurance
C. Use a different supplier
D. Create buffer in the form of management reserve or extra time to schedule

67. Buying insurance is a form of

A. Risk reduction
B. Risk absorption
C. Risk sharing
D. Buying insurance

68. The Pentium is _____ bit microprocessor that was introduce by Intel on March 22, 1993 the very fast CPU in the Pentium brand?

A. 16
B. 32
C. 64
D. 128

69. How do you plot multiple graphs in MATLAB?

A. Using the hold function
B. Using the mplot function
C. Using the mstem function
D. It cannot be done in matlab

70. What built-in definition does “I” have?

A. basic imaginary unit
B. index function
C. infinity
D. index variable

71. All of the following are characteristics of a project except?

A. Repeat itself every month
B. Interrelated activities
C. Definite beginning and end
D. Temporary

72. What is the term for a high-speed memory use to store a small amount of dat temporary so that the data can be retrieve quickly?

A. Video memory
B. Scratchpad memory
C. Cache memory
D. Volatile memory

73. What does the Profiler track?

A. Execution time

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B. Errors
C. Command history
D. The value of variables

74. Refers to the process of increasing the electrical properties and/or characteristics of the power line and its associated facilities to accommodate the present and the future requirements of the system or of the grid?
A. Uprating
B. Upgrading
C. Research and development
D. Improvement

75. Color used for traffic and good housekeeping signage?
A. Blue B. Yellow with white C. Black and white D. Red

76. What is the oxidation number of nitrogen in NO?
A.-1 B. +1 C. +3 D. +5

77. How are lines of constant pressure in a fluid related to the force field?
A. They are parallel to the force field.
B. They are perpendicular to the force field.
C. They are at a 45° angle to the force field.
D. They are perpendicular only to the fore of gravity.

78. In preparing a metallographic iron specimen, the grain boundaries are most visible by which of the following steps?
A. Grinding the sample with silicon carbide abrasive
B. Polishing the sample with A1203
C. Mounting the sample in an epoxy resin mold
D. Etching the sample in a 2% solution of nitric acid in alcohol*

79. In RA 7920, "commission" refers to
A. PRC B. IIEE C. PRBEE D. DOLE

80. In RA 7920, which of the following is not considered as a practice of electrical engineering?
A. Manufacturing of electrical equipment
B. Sales and distribution of electrical equipment
C. Preparing electrical plans
D. Teaching in secondary school

81. Cross linking of natural rubber is called?
A. Sulfurization B. Polymerization C. Vulcanization D. Rubberization

82. Specific atom of an isotope is called
A. Nuclide B. Electron C. Fermion D. Proton

83. IEEE 1284 is a standard communication that defines
A. Wireless communication
B. Parallel communication
C. Wired communication
D. Serial communication

84. A network of private computers infected with malicious software and controlled as a group without the owner's knowledge, e.g., to send spam messages.
A. Phishing B. Hacking C. Botnet D. Trojan horse.

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85.What type of curve is generated by a point that moves in uniform circular motion about an axis, while travelling with a constant speed v, parallel to the axis?
A. A cycloid B. An epicycloid C. A hypocycloid D. A helix

86. Which of the following flow meters' measure(s) the average fluid velocity rather than a point or local velocity in a pipe?
I. Venturi meter
II. Pitot tube
III. Impact tube
IV. Orifice meter
V. Hot-wire anemometer
A. I only B. II only C. I and IV D. II and V

87. Which of the following thermodynamic relations is INCORRECT?
A. TdS=du+pdV
B. Tds=dH-Vdp
C. U=Q-W
D. H=U+ PV

88.Engineers can become good managers only through.
A. Experience
B. Effective career planning
C. Taking master's degree
D. Trainings

89.For which of the following is the Program Evaluation and Review Technique (PERT) NOT used?
A. Construction of projects
B. Computer programming assignments
C. Preparation of bids and proposal
D. Queueing problems

90.When an engineer enters management, what is the most likely problem that he finds difficult to acquire?
A. Learning to trust others.
B. Learning how to work through others
C. Learning how to take satisfaction in the work of others
D. All of the above

91.What is the most essential attribute of a project manager?
A. Leadership B. Charisma C. Communication skill D. Knowledge

92. Is a software that allows users to have a fast and flexible access to information in large documents, constructing associations among data items
A. Voice mail B. Internet C. E-mail D. Hypertext

93. Which of the following part is commonly used in order to speed up the transfer of the computer data?
A. Parallel port B. Serial port C. Video port D. Modem port

94. An instruction that causes data in the accumulator to be moved to the memory or a peripheral register
A. FETCH B. STORE C. LOOP D. LOAD

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95. Which work mode is a non-equilibrium work mode?
A. Compressing a string
B. Transmitting torque with a rotating shaft
C. Energizing an electrical resistor
D. Compressing gas in a cylinder

96. Which of the following computer program is NOT considered open-source software?
A. Linux B. Windows 7 C. OpenOffice.org D. Apach

97. Protection of workers from impact penetration from falling and flying objects. blows, and from limited electric shock and burns shall be provided where there is reasonable probability of exposure to such hazards.
A. Hard Hat B. Toe Board C. Harness D. Belt

98. The three-moment equation may be used to analyze
A. An axially end-loaded beam
B. A beam loaded at the third points
C. A continuous beam
D. A step tapered column

99. What is the form of the moment diagram of a beam supported at both ends and carrying a uniformly distributed load?
A. Triangle B. Hyperbola C. Parabola D. Rectangle

100. What is the weight percentage of engineering sciences and allied subjects?
A. 40 B. 35 C. 30 D. 25

100. Which of the following is a ground for disqualification for the applicant for REE licensure examination?
A. Has not been finally convicted by the court of an offense involving moral turpitude
B. Good reputation with low moral values
C. Citizen of the Philippines
D. At least twenty-one (21) years of age

101. In a combustion chamber, fuel is burned to raise the temperature of the medium prior to production of work through an expansion process. The heat generated during the combustion reaction is the
A. Heat of formation
B. Specific heat
C. Heat of vaporization
D. Heat of reaction

102. How is material purchase variance defined?
A. (quantity purchased) (actual price) (quantity purchased) (standard price)
B. (quantity issued standard quantity) (standard price)
C. (actual price standard price) (quantity used)
D. (quantity purchased quantity used) (actual price)

103. Which of the following is not a requirement for one to be a member of BEE?
A. Person with final conviction by the court of an offense involving moral turpitude
B. A natural born Filipino citizen and a resident of the Philippines for at least five consecutive years

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C. At least thirty-five years of age, of proven integrity with high moral values in his personal as well as his professional conduct
D. A professional electrical engineer with a valid certificate of registration and a valid professional license duly qualified to practice electrical engineering in the Philippines

104. The severity of an electric shock depends on
A. Path of current flow
B. Amount of current flow
C. Duration of current flow
D. All of the above

105. What is the process used to eliminate redundancy in database systems?
A. Normalization B. Abnormalization C. Hashing D. Thrashing

106. Which are NOT included in the duties and responsibilities of PRBEE?
A. Supervise and regulate the practice of electrical engineering in the Philippines
B. Issue special permit to individual foreign electrical engineer and electricians for specific projects
C. All of these
D. Perform such other functions and duties as may be necessary to implement effectivity of this Act

107. What color is used to designate caution limited to warning against starting, use of, or movement of equipment which are under repair or being worked upon?
A. Blue B. Orange C. Red D. Yellow

108. Equation for converting temperature from Celsius to Fahrenheit.
A. Tf 9/5 Tc + 32
B. Tf 5/9 Tc + 32
C. Tf 9/5 Tc 32
D. Tf 5/9 Tc 32

109. The test used to determine the chemical composition of coal
A. Proximate analysis
B. Ultimate analysis
C. Orsat analysis
D. Stoichiometry.

110. What is true about current flow in multiple paths? It will only flow in the least value of resistance path
A. TRUE
B. FALSE
C. Either TRUE or FALSE
D. Neither True nor FALSE

111. When one electrical circuit controls another circuit through a relay, the first circuit is called a
A. Primary circuit
B. Remote-control circuit
C. Signal circuit
D. Controller

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112. Which of the following is not a hazardous practice?
A. Use of proper tools/equipment
B. Using improvised tools
C. Servicing equipment in motion
D. Operating without authority

113. Also known as PRC Renewal Law
A. RA 8189 B. RA 8981 C. RA 7920 D. RA 184

114. Which of the following are NOT examples of nonvolatile memory in the computer system?
A. Flash drives.
B. RAMS
C. Optical disk
D. Hard drives

115. Which of the following fluorescent luminaires CANNOT be directly mounted?
A. Framing members B. Concrete ceiling C. Plasters D. Plywood ceiling

116. Which of the following does not threaten the operation of a computer?
A. Trojan Horse B. Worm C. Firewall D. Hacking

117. Protection of workers form impact penetration from falling and flying objects, blows, and from limited electric shock and burns shall be provided where there is reasonable probability of exposure of such hazards.
A. Hard Hat B. Toe Board C. Harness D. Belt

118. Which of the following is an example of an integrity violation of e-commerce security?
A. website is not actually operated by the entity the customer behaves it to be
B. A merchant uses customer information in a manner not intended by the customer
C. A customer denies that he or she is the person who placed the order
D. An unauthorized person intercepts an online communication and changes the content

119. Refers to burning of waste and other materials, excavation, quarrying, construction, and other activities of the same nature and character, which are performed partially or wholly inside the power line corridor which threaten or endanger the continuous and uninterrupted. electricity; conveyance of
A. Hazardous improvements
B. Hazardous activities
C. Power line obstruction
D. Unsafe Act

120. "In the REE examination of Failed Subjects in the examination for Registered Electrical Engineer, an applicant times, only the subject/s in which he has obtained a grade below fifty shall be allowed to retake
A. three B. no limit C. four D. six

121. Which of the following is NOT a power and duties of the Board of Electrical Engineering?
A. Issue special permits to any foreign electrical engineers and electricians for any projects and for an indefinite duration of time
B. Prosecute or institute criminal action against any violator of the Act and/or

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the rules and regulations of the Board
C. Determine and evaluate the qualifications of the applicants for registration with or without licensure examinations and special permits for
D. Prescribe, amend or revise the requirements for professional electrical engineers and subjects in the licensure examination for registered electrical engineers and registered master electricians and their relative weights, subject to the approval of the Commission

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ANSWER KEY

1.D
2.D
3.B
4.C
5.B
6.C
7.B
8.D
9.A
10.A
11.C
12.C
13.B
14.D
15.B
16.B
17.C
18.A
19.C
20.C
21.C
22.A
23.A
24.C
25.B
26.C
27.B
28.C
29.D
30.A
31.D
32.D
33.B
34.A
35.C
36.C
37.C
38.C
39.C
40.B
41.D
42.B
43.B
44.D
45.C
46.D
47.C
48.D
49.D
50.D
51.C
52.D
53.B
54.D
55.D
56.B
57.A
58.C
59.D
60.D
61.D
62.D
63.D
64.C
65.C
66.D
67.C
68.B
69.A
70.A
71.A
72.B
73.A
74.A
75.C
76.D
77.B
78.D
79.A
80.D
81.C
82.A
83.B
84.C
85.D
86.C
87.C
88.B
89.D
90.D
91.A
92.D
93.A
94.B
95.C
96.B
97.A
98.C
99.C
100.B
101.D
102.A
103.A
104.D
105.A
106.B
107.A
108.A
109.B
110.B
111.B
112.A
113.B
114.B
115.A
116.C
117.A
118.D
119.B
120.B
121.A

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